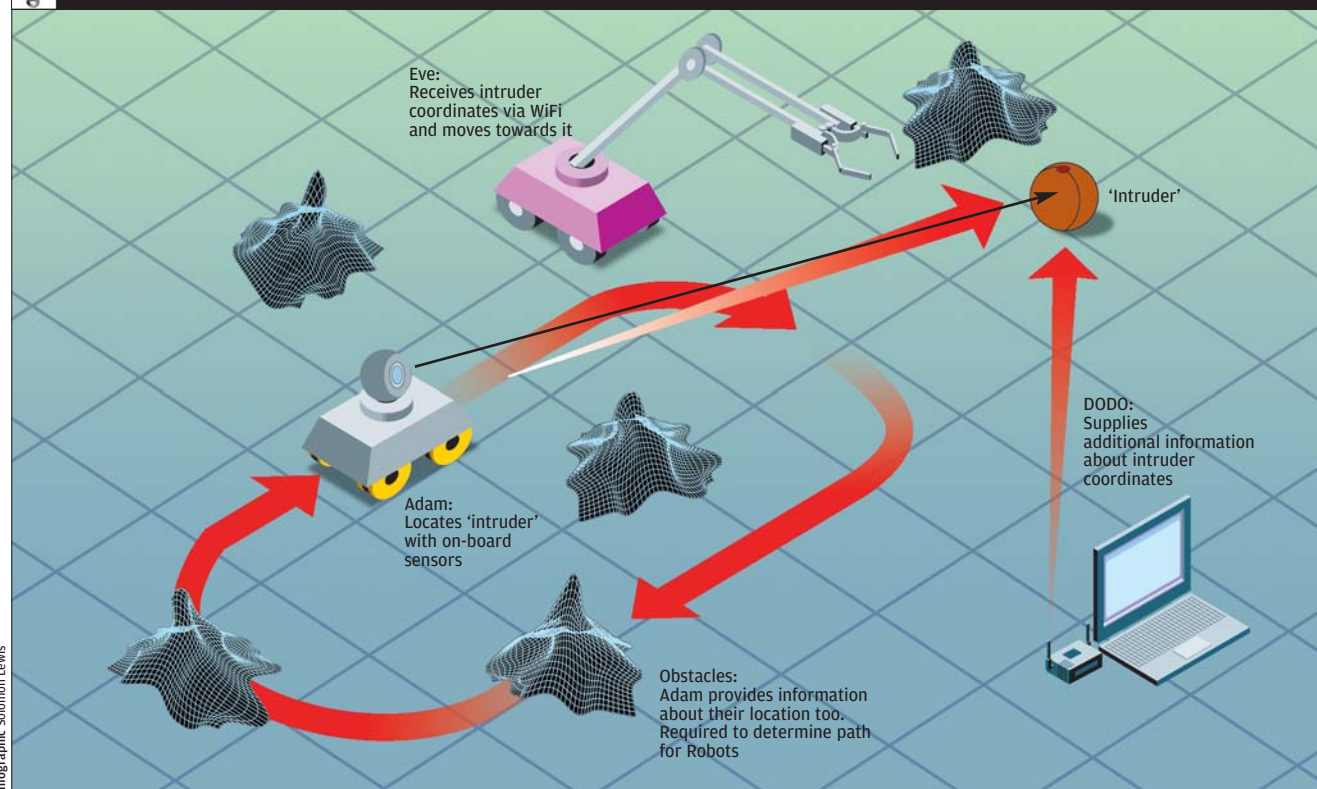


? How Adam And Even Work



Infographic Solomon Lewis

things Linux in this project. As he points out, “We chose Linux for its technical capabilities. Open Source always means greater functionality.” But did they write their code from scratch? Mostly, says Mrinal; only a few scraps were borrowed, such as the module to control the video camera.

Before COMRADE, Sharad was involved in audio processing and DSP experiments. He was the brain behind the OpenGL GUI for the monitoring node. The resident expert in troubleshooting routing problems, he also overcame

petty issues with a rather troublesome WiFi driver, which refused to support multicasting.

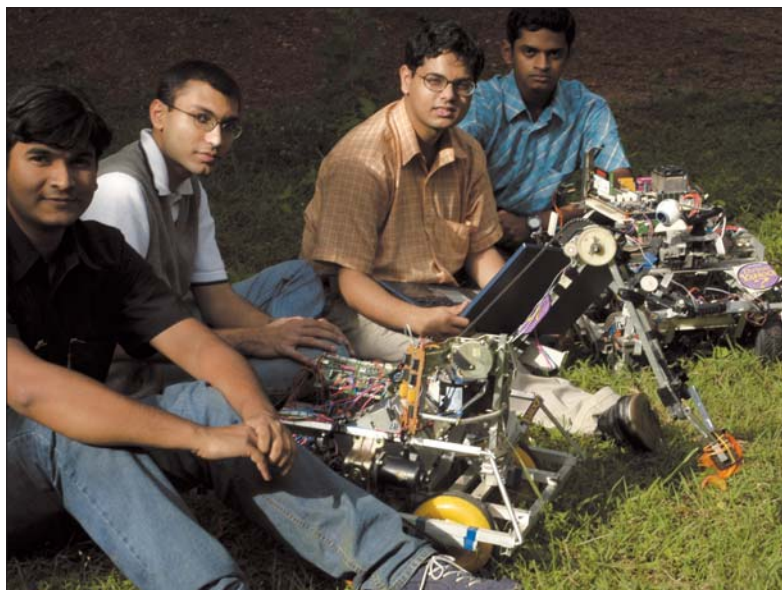
Das was involved in robot construction and architecture—the most critical part of the project. In between solving problems in mechanics, he also found time to team up with Mrinal to design the motor-controller circuit-boards and write the code needed to run them. “The main circuit panel for the smaller motor control boards was built from scratch,” says Mrinal. Das also designed the interface between the lower-level microcontroller-based actuator array and the CPU.

COMRADE’s predecessors—3LER-3, Omnibot, ROAMER and Dexter, were essentially SONAR-based robots that found their way about with the help of sound. Abhishek wrote IRIS, which helps COMRADEs see via video cameras.

He also played a critical role in designing the robot co-operation architecture, and authored BorgCode, a script that has since become the language of the COMRADEs. Moreover, he designed the final execution environment, which integrated IRIS, the communication system and the control module into a seamless whole.

“We made the two robots with material gathered from scrounge markets and garage junk”

“The hardware wasn’t very sophisticated. The image-processing software had to take that into account,” says Abhishek, creator of IRIS, which is the monitoring node software for Project



THE SIX COMRADES: Clockwise from left: JD, Sharad, Sen Gupta, Mrinal, Adam and Eve